Orthogonality

Goals

1. Orthogonal and independence
2. Orthogonalize a basis of a vector (sub)space with Gram-Schmidt orthogonalization algorithm
3. Orthogonal diagonalization and applications

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Dot product, length

Two vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix}$ in $\mathbb{R}^n$

Dot product

Their **dot product** is

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + \cdots + x_n y_n$$

which is matrix product $\mathbf{x}^T \mathbf{y}$.

Length of a vector

Length of $\mathbf{x}$ is

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}$$

Properties of dot product and length

Let $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ be vectors in $\mathbb{R}^n$. Then

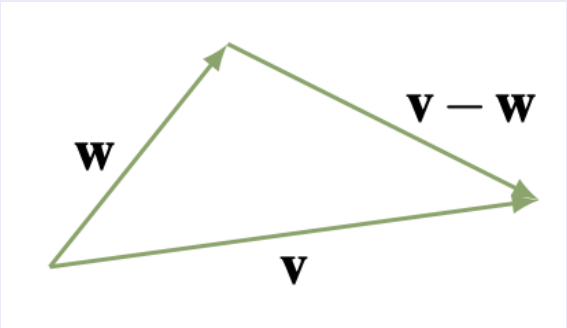
1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$
2. $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$
3. $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$ for all scalar $a \in \mathbb{R}$
4. $\|\mathbf{x}\| = \mathbf{x} \cdot \mathbf{x}$
5. $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$
6. $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$ for all scalar $a \in \mathbb{R}$

Distance

Definition (Euclidean distance)

Distance between two vector $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ is

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$



Properties

Let $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$ be three vectors in $\mathbb{R}^n$ then

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$
2. $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$
3. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
4. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$

Exercise

If $\|\mathbf{x}\| = 3$, $\|\mathbf{y}\| = 1$ and $\mathbf{x} \cdot \mathbf{y} = -2$, compute

1. $(3\mathbf{x} - \mathbf{y}(-\mathbf{x} + 2\mathbf{y}))$
2. $(\mathbf{x} - 2\mathbf{y})(3\mathbf{x} + 5\mathbf{y})$
3. $\|3\mathbf{x} - 5\mathbf{y}\|$
4. $\|2\mathbf{x} + 7\mathbf{y}\|$

Orthogonal and Orthonormal Sets

Orthogonal

Two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ are **orthogonal** if $\mathbf{x} \cdot \mathbf{y} = 0$

Orthogonal sets

A set of vectors $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$ in $\mathbb{R}^n$ is called **orthogonal set** if

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0 \;\; \forall i \neq j \qquad \text{and} \qquad \mathbf{x}_i \neq 0 \;\; \forall i$$

A set of vectors $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$ in $\mathbb{R}^n$ is called **orthonormal** is if is orthogonal and each $\mathbf{x}_i$ is a unite vector, that is

$$\|\mathbf{x}_i\| = 1 \;\; \forall i$$

Example

The standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is an orthonormal set in $\mathbb{R}^n$

Example

If $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k\}$ is an orthogonal set then so is $\{a_1\mathbf{x}_1, a_2\mathbf{x}_2, \cdots, a_k\mathbf{x}_k\}$ for all nonzero scarlars a_i

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Normalizing an Orthogonal Set

If $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k\}$ is an orthogonal set then $\left\{\frac{1}{\|\mathbf{x}_1\|}\mathbf{x}_1, \frac{1}{\|\mathbf{x}_2\|}\mathbf{x}_2, \cdots, \frac{1}{\|\mathbf{x}_k\|}\mathbf{x}_k\right\}$ is an orthogonal set is an orthonormal set.

Example

If $\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$ and $\mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ then $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$ is an orthogonal set in $\mathbb{R}^4$ because

$$\mathbf{f}_1 \cdot \mathbf{f}_2 = \mathbf{f}_2 \cdot \mathbf{f}_3 = \mathbf{f}_3 \cdot \mathbf{f}_1 = 0$$

After normalizing, the orthonomal set is

$$\left\{ \frac{1}{2}\mathbf{f}_1, \frac{1}{\sqrt{6}}\mathbf{f}_2, \frac{1}{\sqrt{2}}\mathbf{f}_3 \right\}$$

Exercise

Verify that the set of vectors

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is orthogonal.
Normalizing this set to obtain an orthonormal set.

Exercise

Find all $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$ such that the set

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \right\}$$

is orthogonal.

Orthogonality implies Linear Independence

Suppose that $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k\}$ is an orthogonal set. Consider

$$t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k = \mathbf{0}$$

Multiplying both sides by $\mathbf{x}_1$, we have

$$t_1\|\mathbf{x}_1\|^2 + t_2\mathbf{x}_2\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k\mathbf{x}_1 = \mathbf{0}$$

It is clear that $\mathbf{x}_2\mathbf{x}_1 = \cdots = \mathbf{x}_k\mathbf{x}_1 = 0$. Hence

$$t_1 \underbrace{\|\mathbf{x}_1\|^2}_{\neq 0} = 0 \Rightarrow t_1 = 0$$

Similar $t_2 = \cdots = t_k = 0$. Hence $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k\}$ is linearly independent

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Similar $t_2 = \cdots = t_k = 0$. Hence $\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k\}$ is linearly independent

Theorem

Every orthogonal set in $\mathbb{R}^n$ is linearly independent

Theorem (Expansion Theorem)

If $\boldsymbol{f}_1, \boldsymbol{f}_2, \ldots, \boldsymbol{f}_m$ is an orthogonal basis of a subspace U of $\mathbb{R}^n$ then for any vector $\boldsymbol{x} \in U$, we have

$$\boldsymbol{x} = \left(\frac{\boldsymbol{x} \cdot \boldsymbol{f}_1}{\|\boldsymbol{f}_1\|^2}\right) \boldsymbol{f}_1 + \left(\frac{\boldsymbol{x} \cdot \boldsymbol{f}_2}{\|\boldsymbol{f}_2\|^2}\right) \boldsymbol{f}_2 + \cdots + \left(\frac{\boldsymbol{x} \cdot \boldsymbol{f}_m}{\|\boldsymbol{f}_m\|^2}\right) \boldsymbol{f}_m$$

The expansion in of $\mathbf{x}$ as a linear combination of the orthogonal basis $\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_m$ is called the **Fourier expansion** of $\mathbf{x}$, and the coefficients $t_i = \frac{\mathbf{x} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$ are called the **Fourier coefficients**.

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Proof.

Suppose that

$$\mathbf{x} = t_1\mathbf{f}_1 + \cdots + t_m\mathbf{f}_m$$

then

$$\mathbf{x} \cdot \mathbf{f}_1 = t_1\mathbf{f}_1 \cdot \mathbf{f}_1 = t_1\|\mathbf{f}_1\|^2$$

So $t_1 = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}$. Similar we have $t_i = \frac{\mathbf{x} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$ for $i = 2, \ldots, m$. □

Example

Let $U = span(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3)$ where $\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$ and $\mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. We have $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$ is an orthogonal set and then is a basis of U. For all vector $\mathbf{x} = (a, b, c, d)$ in U, it can be expanded as a linear combination of $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$ with Fourrier coefficients

$$t_1 = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} = \frac{1}{4}(a + b + c - d)$$
$$t_2 = \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} = \frac{1}{6}(a + c + 2d)$$
$$t_2 = \frac{\mathbf{x} \cdot \mathbf{f}_2}{\|\mathbf{f}_1\|^2} = \frac{1}{2}(-a + c)$$

That is

$$\mathbf{x} = t_1\mathbf{f}_1 + t_2\mathbf{f}_3 + t_3\mathbf{f}_3 = \dots$$

Exercise

Write $\mathbf{x} = \begin{bmatrix} 13 \\ -20 \\ 15 \end{bmatrix}$, as a linear combination of the orthogonal basis of the subspace $U = span \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$

If $\{\mathbf{v}_1, ..., \mathbf{v}_m\}$ is linearly independent, and if $\mathbf{v}_{m+1}$ is not in $span\{\mathbf{v}_1, ..., \mathbf{v}_m\}$, then $\{\mathbf{v}_1, ..., \mathbf{v}_m, \mathbf{v}_{m+1}\}$ is linear independent

Orthogonal Lemma

Let $\{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be an orthogonal set in $\mathbb{R}^n$. Given $\mathbf{x} \in \mathbb{R}^n$, write

$$\mathbf{f}_m = \mathbf{x} - \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|}^2 \mathbf{f}_1 - \cdots - \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|}^2 \mathbf{f}_m$$

then

1. $\mathbf{f}_m \cdot \mathbf{f}_i = 0$ for $k = 1, \cdots m$
2. If $\mathbf{x} \notin span\{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ then $\mathbf{f}_m \neq \mathbf{0}$ and $\{\mathbf{f}_1, \ldots, \mathbf{f}_m, \mathbf{f}_{m+1}\}$ be an orthogonal set

One of the important consequence of orthogonal lemma is an extension for orthogonal sets to a orthogonal basis of a subspace

Theorem

Let U be a subspace in $\mathbb{R}^n$

1. Every orthogonal set in U is a subset of an orthogonal basis of U
2. U has an orthogonal basis

The second consequence of the orthogonal lemma is a procedure by which any basis of a subspace U of $\mathbb{R}^n$ can be systematically modified to yield an orthogonal basis of U

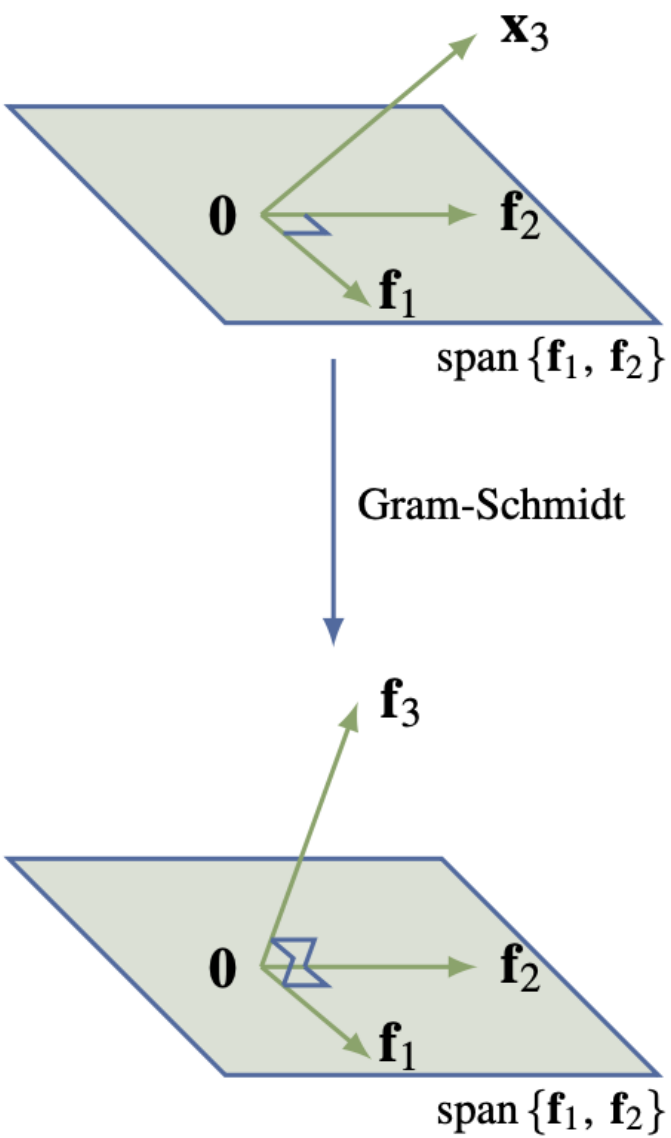
Gram-Schmidt Orthogonalization Algorithm

If $\{\mathbf{x}_1, \cdots, \mathbf{x}_m\}$ is any basis of a subspace U of $\mathbb{R}^n$, construct $\mathbf{f}_1, \mathbf{f}_2, \cdots, \mathbf{f}_m$ in U as following

$$\begin{aligned}
\mathbf{f}_1 &= \mathbf{x}_1 \\
\mathbf{f}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\mathbf{f}_1 \\
&\vdots \\
\mathbf{f}_k &= \mathbf{x}_k - \frac{\mathbf{x}_k \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\mathbf{f}_1 - \frac{\mathbf{x}_k \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2}\mathbf{f}_2 - \frac{\mathbf{x}_k \cdot \mathbf{f}_{k-1}}{\|\mathbf{f}_{k-1}\|^2}\mathbf{f}_{k-1}
\end{aligned}$$

for $k = 2, \ldots, m$. Then

1. $\mathbf{f}_1, \mathbf{f}_2, \cdots, \mathbf{f}_m$ is an orthogonal basis of U
2. $span(\mathbf{f}_1, \cdots, \mathbf{f}_k) = span(\mathbf{x}_1, \cdots, \mathbf{x}_k)$ for all $k = 1, \ldots, m$


$\mathbf{x}_3$
$\mathbf{0}$
$\mathbf{f}_2$
$\mathbf{f}_1$
span $\{\mathbf{f}_1, \mathbf{f}_2\}$
Gram-Schmidt
$\mathbf{f}_3$
$\mathbf{0}$
$\mathbf{f}_2$
$\mathbf{f}_1$
span $\{\mathbf{f}_1, \mathbf{f}_2\}$

Example

Find an orthogonal basis of $U = span\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ with

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Solution

Observe that $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ are independent. The algorithm gives

$$\mathbf{f}_1 = \mathbf{x}_1$$

$$\mathbf{f}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\mathbf{f}_1 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix}^T - \frac{4}{4}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\mathbf{f}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\mathbf{f}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2}\mathbf{f}_2 = \mathbf{x}_3 - \frac{0}{4}\mathbf{f}_1 - \frac{3}{10}\mathbf{f}_2 = \frac{1}{10}\begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix}$$

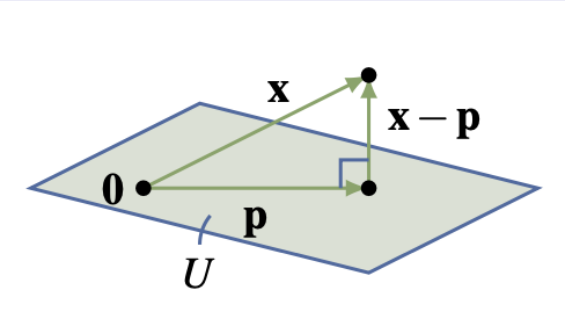
Hence $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3\}$ is an orthogonal basis.

Remark

The orthogonal property does not change it a vector in basis is multiplied by a nonzero. It may be convenient to eliminate fractions and use $\{\mathbf{f}_1, \mathbf{f}_2, 10\mathbf{f}_3\}$ as an orthogonal basis of U

Problem motivation

Suppose a point $\mathbf{x}$ and a plane U through the origin in $\mathbb{R}^3$ are given, and we want to find the point $\mathbf{p}$ in the plane that is closest to $\mathbf{x}$. Our geometric intuition assures us that such a point $\mathbf{p}$ exists. In fact, $\mathbf{p}$ must be chosen in such a way that $\mathbf{x} - \mathbf{p}$ is perpendicular to the plane.



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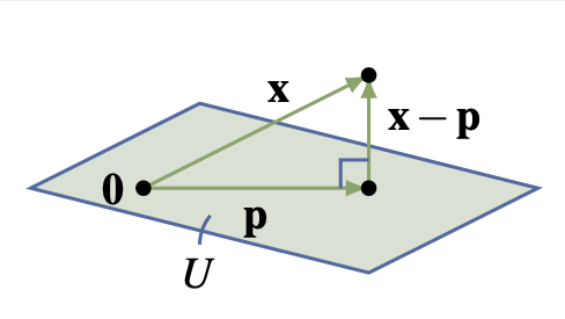


Orthogonal Complement

If U is a subspace of $\mathbb{R}^n$, define the orthogonal complement $U^\perp$ of U (pronounced "U-perp") by

$$U^\perp = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \quad \forall \mathbf{y} \in U\}$$

Properties of the orthogonal complement

Lemma

Let U be a subspace of $\mathbb{R}^n$

1. $U^\perp$ is a subspace of $\mathbb{R}^n$
2. $\{\mathbf{0}^\perp = \mathbb{R}^n\}$ and $(\mathbb{R}^n)^\perp = \{\mathbf{0}\}$
3. If $U = span\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ then
   $U^\perp = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{x}_i = 0 \quad \text{for } i = 1, \ldots, m\}$

Example

Find $U^{\perp}$ if $U = span\left\{\left[1,-2,2,0\right]^T, \left[1,0,-2,3\right]^T\right\}$ in $\mathbb{R}^4$.

Solution

$\boldsymbol{x} = \left[x,y,z,w\right]^T$ *is in* $U^{\perp}$ *if and only if it is orthogonal to both* $\left[1,-2,2,0\right]^T$ *and* $\left[1,0,-2,3\right]^T$*, that is*

$$x - y + 2z = 0 \tag{1}$$

$$x - 2z + 3w = 0 \tag{2}$$

Gaussian elimimation gives $U^{\perp} = span\left\{\left[2,4,1,0\right]^T, \left[3,3,0,-1\right]^T\right\}$

Projection onto a Suspace of $\mathbb{R}^n$

Let U be a subspace in $\mathbb{R}^n$ with orthogonal basis $\{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$. The vector $\mathbf{x}$ in $\mathbb{R}^n$ is defined by

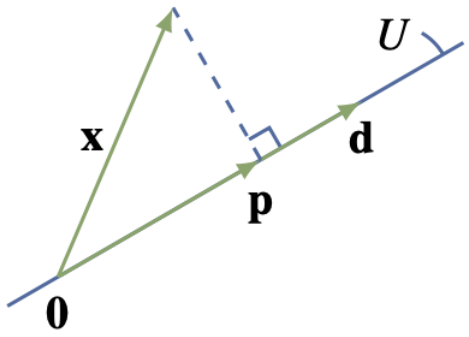
$$\text{proj}_U \mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\mathbf{f}_1 + \cdots + \frac{\mathbf{x} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2}\mathbf{f}_m$$

is called the **orthogonal projection** of $\mathbf{x}$ on U.

For the zero subspace $U = \{\mathbf{0}\}$, we define

$$\text{proj}_{\{\mathbf{0}\}} \mathbf{x} = \mathbf{0}$$

Projection onto a line in $\mathbb{R}^3$



Consider vectors $\mathbf{x}$ and $\mathbf{d} \neq \mathbf{0}$ in $\mathbb{R}^3$. The projection $\mathbf{p} = \text{proj}_{\mathbf{d}}\mathbf{x}$ is defined as

$$\mathbf{p} = \text{proj}_{\mathbf{d}}\mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{d}}{\|\mathbf{d}\|^2}\mathbf{d}$$

where the dotted line is the error $e = \mathbf{x} - \mathbf{p}$ which is orthogonal (perpendicular) to $\mathbf{d}$.

Example

Find the projection of $\mathbf{u} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ on $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and express $\mathbf{x}$ as $\mathbf{u}_1 + \mathbf{u}_2$ where $\mathbf{u}_1$ is parallel to $\mathbf{d}$ and $\mathbf{u}_2$ is orthogonal to $\mathbf{d}$.

Example

Find the projection of $\mathbf{u} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$ on $\mathbf{d} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and express $\mathbf{x}$ as $\mathbf{u}_1 + \mathbf{u}_2$ where $\mathbf{u}_1$ is parallel to $\mathbf{d}$ and $\mathbf{u}_2$ is orthogonal to $\mathbf{d}$.

Solution

The projection $\boldsymbol{u}_1$ *of* $\boldsymbol{u}$ *on* $\boldsymbol{d}$ *is*

$$\boldsymbol{u}_1 = proj_{\boldsymbol{d}}\boldsymbol{u} = \frac{\boldsymbol{u} \cdot \boldsymbol{d}}{\|\boldsymbol{d}\|}\boldsymbol{d} = \frac{2+3+3}{1^2+(-1)^2+3^2}\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} = \frac{8}{11}\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

Hence $\boldsymbol{u}_2 = \boldsymbol{x} - \boldsymbol{u}_1 = \frac{1}{11}\begin{bmatrix} 14 \\ -25 \\ -13 \end{bmatrix}$ *is orthogonal to* $\boldsymbol{d}$.

Theorem (Projection Theorem)

If U is a subspace in $\mathbb{R}^n$ and denote $\boldsymbol{p} = proj_U \boldsymbol{x}$ then

1. $\boldsymbol{p} \in U$ *and* $\boldsymbol{x} - \boldsymbol{p} \in U^\perp$
2. $\boldsymbol{p}$ *is the vector in* U *closest to* $\boldsymbol{x}$ *in the sense that*

$$\|\boldsymbol{x} - \boldsymbol{p}\| < \|\boldsymbol{x} - \boldsymbol{y}\| \quad \forall \boldsymbol{y} \in U, \boldsymbol{y} \neq \boldsymbol{p}$$

Example

Let $U = span\{\mathbf{x}_1, \mathbf{x}_2\}$ where $\mathbf{x}_1 = \begin{bmatrix}1, 1, 0, 1\end{bmatrix}^T$ and $\mathbf{x}_2 = \begin{bmatrix}0, 1, 1, 2\end{bmatrix}^T$. If $\mathbf{x} = \begin{bmatrix}3, -1, 0, 2\end{bmatrix}^T$. Find the vector in U closest to $\mathbf{x}$ and express $\mathbf{x}$ as the sum of a vector in U and a vector orthogonal to U.

Solution

$\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$ are independent but not orthogonal. Gram-Schmidt algorithm gives an orthogonal basis $\{\boldsymbol{f}_1, \boldsymbol{f}_2\}$ of U where $\boldsymbol{f}_1 = \boldsymbol{x}_1 = \left[1, 1, 0, 1\right]^T$ and

$\boldsymbol{f}_2 = \boldsymbol{x}_2 - \frac{\boldsymbol{x}_2 \cdot \boldsymbol{f}_1}{\|\boldsymbol{f}_1\|}^2 \boldsymbol{f}_1 = \boldsymbol{x}_2 = \frac{3}{3}\boldsymbol{f}_1 \left[-1, 0, 1, 1\right]^T$

Compute projection using orthogonal basis $(\boldsymbol{f}_1, \boldsymbol{f}_2)$

$$\boldsymbol{p} = proj_U \boldsymbol{x} = \frac{\boldsymbol{x} \cdot \boldsymbol{f}_1}{\|\boldsymbol{f}_1\|}^2 \boldsymbol{f}_1 + \frac{\boldsymbol{x} \cdot \boldsymbol{f}_2}{\|\boldsymbol{f}_2\|}^2 \boldsymbol{f}_2 = \frac{4}{3}\boldsymbol{f}_1 + \frac{-1}{3}\boldsymbol{f}_2 = \frac{1}{3}\left[{}^T 5, 4, -1, 3\right]$$

Thus $\boldsymbol{p}$ is the vector in U closest to $\boldsymbol{x}$ and $\boldsymbol{x} - \boldsymbol{p} = \frac{1}{3}\left[4, -7, 1, 3\right]^T$ is orthogonal to every vector in U. The decomposition of $\boldsymbol{x}$ is

$$\boldsymbol{x} = \boldsymbol{p} + (\boldsymbol{x} - \boldsymbol{p}) = \ldots$$

Projection Onto a Subspace

Let $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ be a basis of a subspace U in $\mathbb{R}^m$

Problem: find the combination $\mathbf{p} = \hat{x}_1\mathbf{a}_1 + \cdots + \hat{x}_n\mathbf{a}_n$ closest to a vector $\mathbf{b} \in U$.

We need to find $\mathbf{p} = A\hat{\mathbf{x}}$ - **the projection** in the column space of $A = \begin{bmatrix} \mathbf{a}_1 & \ldots & \mathbf{a}_n \end{bmatrix}$ that closest to $\mathbf{b}$. The **error vector** $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ is perpenticular to the space

$$\begin{cases} \mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \\ \vdots \\ \mathbf{a}_2^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \end{cases} \quad \text{or} \quad \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{b} - A\hat{\mathbf{x}} \end{bmatrix} = \mathbf{0} \Longleftrightarrow A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$$

Hence

$$A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$$

If **a**'s are linear independent then $A^T A$ is symmetric and invertible then

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

So

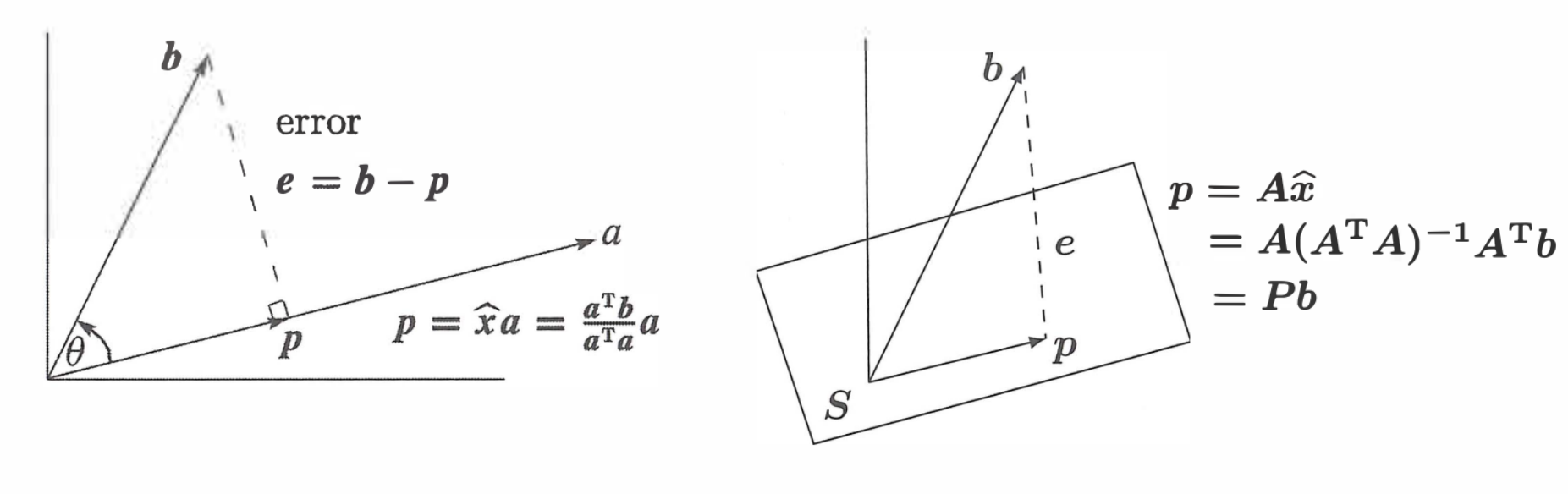
$$\mathbf{p} = A\hat{\mathbf{x}} = A(A^T A)^{-1} A^T \mathbf{b}$$

The matrix

$$P = A(A^T A)^{-1} A^T$$

is the **projection matrix** such that $\mathbf{p} = \mathrm{P}\mathbf{b}$

Visualization of projection onto a line and onto $S =$ column space of A



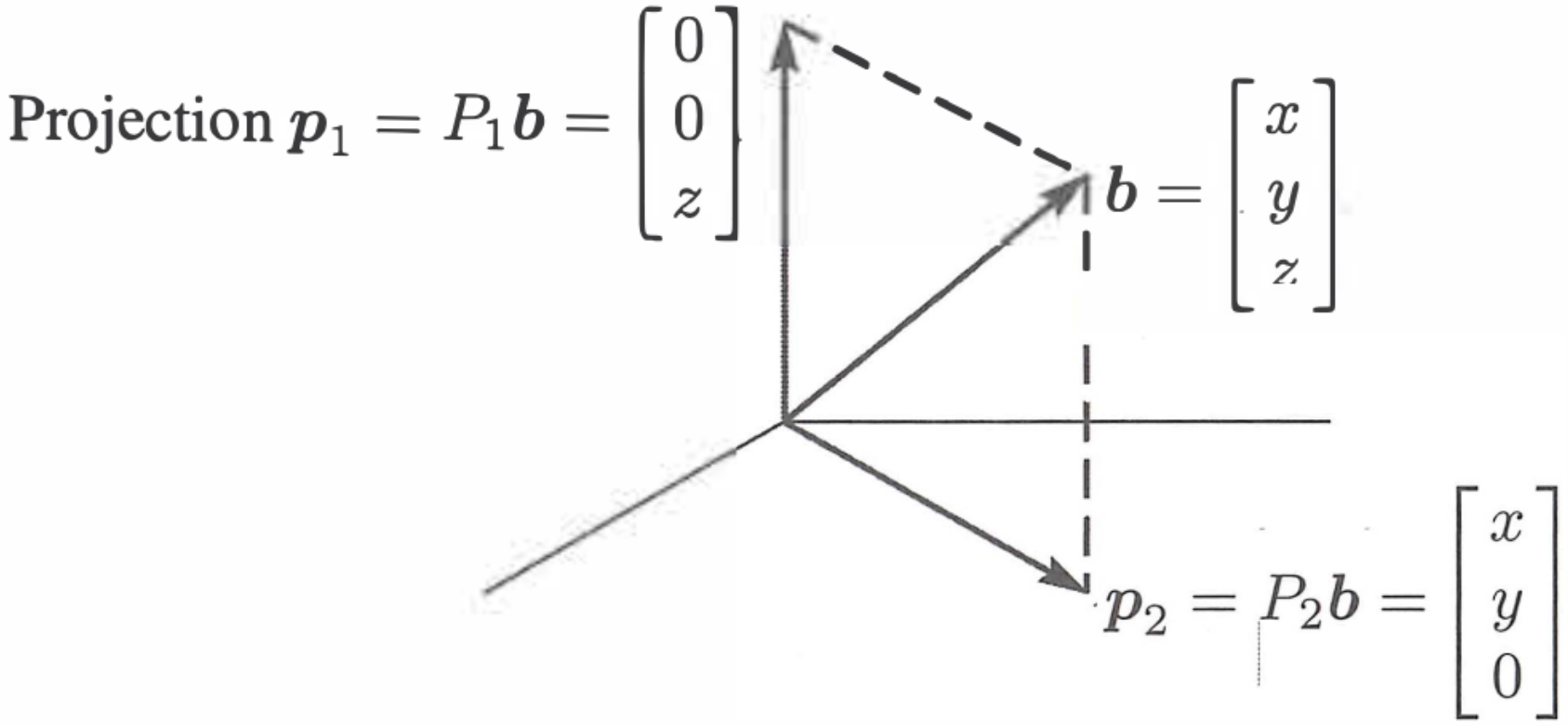
Let θ be the angle between **b** and the line go through vector **b** then the projection **p** of **b** on the line has the length

$$\|\mathbf{b}\| = \frac{\|\mathbf{a}^T\|\|\mathbf{b}\|\cos\theta}{\|\mathbf{a}\|^2}\|\mathbf{a}\| = \|\mathbf{b}\|\cos\theta$$

and the length of error

$$\|\mathbf{e}\| = \|\mathbf{b}\|\sin\theta$$

Projection in $\mathbb{R}^3$



$\mathbf{p}_1$ is projection of $\mathbf{b}$ on the line Oz - axis $= span\{\mathbf{e}_3\}$ and $\mathbf{p}_2$ is the projection of $\mathbf{b}$ onto the plane $Oxy = span\{\mathbf{e}_1, \mathbf{e}_2\}$

Projection on Oz

Matrix of subspace

$$A = \begin{bmatrix} \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So $A^T A = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 1$. Hence projection matrix

$$P_1 = A(A^T A)^{-1} A^T = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} 1 \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The projection of $\mathbf{b} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is $\mathbf{p}_1 = P_1 \mathbf{b} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix}$

Projection onto Oxy

Matrix of subspace

$$A = \left[\mathbf{e}_1, \mathbf{e}_2\right] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

So $A^T A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Hence projection matrix

$$P_2 = A(A^T A)^{-1} A^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The projection of $\mathbf{b} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is $\mathbf{p}_2 = P_2 \mathbf{b} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$

Exericise

Let $U = span\{\mathbf{x}_1, \mathbf{x}_2\}$ where $\mathbf{x}_1 = \begin{bmatrix}1, 1, 0, 1\end{bmatrix}^T$ and $\mathbf{x}_2 = \begin{bmatrix}0, 1, 1, 2\end{bmatrix}^T$. Find projection of $\mathbf{b} = \begin{bmatrix}3, -1, 0, 2\end{bmatrix}^T$ on U.

Exercise

Let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}.$$

Find projection of **b** on column space of A.

Application to Least Square Approximation

1. Solving $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$ gives the projection $\mathbf{p} = A\hat{x}$ of $\mathbf{b}$ onto the column space of A
2. When $A\mathbf{x} = \mathbf{b}$ has no solution then $\hat{\mathbf{x}}$ is the "least-square solution": $\|\mathbf{b} - A\hat{\mathbf{x}}\|^2 =$ minimum
3. Setting partial deriavatives of square of length of error $E = \|\mathbf{b} - A\hat{\mathbf{x}}\|^2$ to zero $\left(\frac{\partial E}{\partial x_i} = 0\right)$ also produces $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$

Example

The system of linear equation

$$\begin{cases} 3x - y = 4 \\ x + 2y = 0 \\ 2x + y = 1 \end{cases} \tag{3}$$

has no solution. Find the vector $\hat{\mathbf{x}} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ that best approximates a solution.

Solution

We have $A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix}$ *and* $\boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$. *So*

$$A^T A = \begin{bmatrix} 14 & 1 \\ 1 & 6 \end{bmatrix}$$

is invertible. So the best approximating solution is

$$\hat{\boldsymbol{x}} = (A^T A)^{-1} A^T \boldsymbol{b} = \frac{1}{83} \begin{bmatrix} 87 \\ -56 \end{bmatrix}$$

Exercise

Find a linear function $g = a_0 + a_1 x_1 + a_2 x_2$ that best fits these data

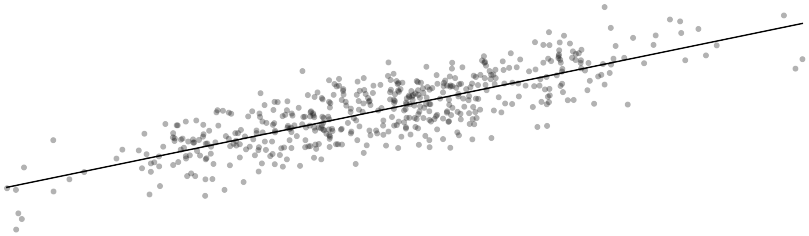
g	x_1	x_2
0.8	5	3
0.8	3	4
0.6	1	5
0.4	2	1

Fitted Line - Linear Regression

To fit points (t_1, b_1), ..., (t_m, b_m) by a straight lines $y = k + ax$, we need to solve

$$\begin{cases} k + at_1 = b_1 \\ \dots \\ k + at_m = b_m \end{cases} \quad \text{or } A\mathbf{x} = \mathbf{b}$$

$$\text{with } A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} k \\ a \end{bmatrix}$$



The coefficient is best fitted with the approximating solution $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$

Example

Find the closest line to the points $(0,6),(1,0)$ and $(2,0)$

Solution

We have $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ *and* $\boldsymbol{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ *The coefficients* a, k *of the fitted line are given by*

$$\begin{bmatrix} k \\ a \end{bmatrix} = \hat{\boldsymbol{x}} = (A^T A)^{-1} A^T \boldsymbol{b} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

So the fitted line is $y = 5 + 3x$

Exercise

Find the least squares approximating line for the data

x	y
1	1
3	2
4	3
6	4
7	5

Fitting by a Parabola

Drop a stone from the Leaning Tower of Pisa by Galileo

Fit heights $b_1, \ldots, b_m$ at times $t_1, \ldots, t_m$ by a parabola $C + Dt + Et^2$

The exact fit are solution to system

$$\begin{cases} C + Dt_1 + Et_1^2 = b_1 \\ \ldots \\ C + Dt_m + Et_m^2 = b_m \end{cases} \Leftrightarrow A\mathbf{x} = \mathbf{b} \text{ with } A = \begin{bmatrix} 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_m & t_m^2 \end{bmatrix}$$

which is generally unsolvable

Least square The closet parabola $C + Dt + Et^2$ chooses $\hat{\mathbf{x}} = \begin{bmatrix} C \\ D \\ E \end{bmatrix}$

satisfy $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$

Example

Find the least squares approximating quadratic $y = a_0 + a_1 x + a_2 x^2$ for the following data points $(-3, 3), (-1, 1), (0, 1), (1, 2), (3, 4)$

Example

Find the least squares approximating quadratic $y = a_0 + a_1 x + a_2 x^2$ for the following data points $(-3, 3), (-1, 1), (0, 1), (1, 2), (3, 4)$

Solution

Consider

$$A = \begin{bmatrix} 1 & -3 & (-3)^2 \\ 1 & -1 & (-1)^2 \\ 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 3 & 3^2 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 2 \\ 4 \end{bmatrix}$$

We have

$$A^T A = \begin{bmatrix} 5 & 0 & 20 \\ 0 & 20 & 0 \\ 20 & 0 & 60 \end{bmatrix}$$

is invertible. So the approximating coefficient is given by

Exercise

Find the least squares approximating quadratic $s = a_0 + a_1 t + a_2 t^2$ for the data

t	1	2	3
s	95	80	56

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   - Projection
2. Orthogonal Diagonalization
3. Singular Value Decomposition (SVD)

Question

An $n \times n$ matrix A is diagonalizable if and only if it has n independent eigenvectors. The matrix P with these eigenvectors as column is a diagonalization matrix for A, that is

$$P^{-1}AP \text{ is diagonal}$$

The really nice bases of $\mathbb{R}^n$ are the orthogonal ones. So **which matrices have an orthogonal basis of eigenvectors?**

Answer this question is a main result in this section: the matrix A is symmetric

Normalize an orthogonal set

Recall that an orthogonal set of vector is orthogonormal if all vector has length of 1. Any orthogonal set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ can be (normalized) converted to an orthogonormal set $\left\{\frac{1}{\|\mathbf{v}_1\|}\mathbf{v}_1, \ldots, \frac{1}{\|\mathbf{v}_k\|}\mathbf{v}_k\right\}$

Orthogonal matrix

Theorem

The following conditions are equivalent for all square matrix P

1. *P is invertible and $P^{-1} = P$*
2. *The rows of P are orthogonormal*
3. *The columns of P are orthogonormal*

Definition

A square matrix P is called an **orthogonormal matrix** if it satisfies one of the above conditions

Example

The rotation matrix $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal for any angle θ

It is not enough that the rows of a matrix A are merely orthogonal for A to be an orthogonal matrix.

Example

The matrix

$$\begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

has orthogonal rows but columns are not orthogonal.

However it the rows are normalized then the resulting matrix

$$\begin{bmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

is orthogonal

Example

if P and Q are orthogonal matrices then PQ and P are also orthogonal

Example

if P and Q are orthogonal matrices then PQ and P are also orthogonal

Solution

Prove that PQ is orthogonal

P and Q are invertible and so is PQ with

$$(PQ)^{-1} = Q^{-1}P^{-1}$$

Because P and Q are orthogonal, we have $P^{-1} = P^T$ and $Q^{-1} = Q^T$. Hence

$$(PQ)^{-1} = Q^T P^T = (PQ)^T$$

Thus PQ is orthogonal

Solution (Cont)

Prove that P^{-1} is orthogonal

It is clear that P^{-1} is invertible with

$$(P^{-1})^{-1} = P$$

Moreover P is orthogonal, so $P^{-1} = P^T$. Hence

$$(P^{-1})^T = (P^T)^T = P$$

Thus

$$(P^{-1})^{-1} = (P^{-1})^T$$

So P^{-1} is orthogonal

Definition (Orthogonally Diagonalizable Matrices)

An $n \times n$ matrix A is said to be **orthogonally diagonalizable** if here exists an orthogonal matrix P such that $P^{-1}AP = P^TAP$ is diagonal

This condition turns out to characterize the symmetric matrices

Theorem (Principal Axes Theorem)

The following conditions are equivalent for a square matrix A

1. *A is an orthogonal set of n eigenvectors*
2. *A is orthogonally diagonalizable*
3. *A is symmetric*

A set of orthonormal eigenvectors of a symmetric matrix A is called a set of **principal axes** for A. The name comes from geometry and this is discussed in application to quadratic form later

Example

Find an orthogonal matrix P such that $P^{-1}AP$ is diagonal where

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

Solution

The characteristic polynomial of A is

$$c_A(x) = \det \begin{bmatrix} 1-x, 0, -1 \\ 0, 1-x, 2 \\ -1, 2, 5-x \end{bmatrix} = -x(x-1)(x-6)$$

Thus the eigenvalues are $\lambda_1 = 0$, $\lambda_2 = 1$ and $\lambda_3 = 6$. The corresponding eigenvectors are

$$\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad \boldsymbol{x}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{x}_3 = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

These vectors are orthogonal so they should be normalized to create an orthogonal diagonalizing matrix

$$P = \begin{bmatrix} \frac{\mathbf{x}_1}{\|\mathbf{x}_1\|} & \frac{\mathbf{x}_2}{\|\mathbf{x}_2\|} & \frac{\mathbf{x}_3}{\|\mathbf{x}_3\|} \end{bmatrix} = \frac{1}{\sqrt{30}} \begin{bmatrix} \sqrt{5} & 2\sqrt{6} & -1 \\ -2\sqrt{5} & \sqrt{6} & 2 \\ \sqrt{5} & 0 & 5 \end{bmatrix}$$

Thus $P^{-1} = P^T$ and

$$P^T A P = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Exercise

Orthogonally diagonalize the symmetric matrix $A = \begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & 4 \\ 2 & 4 & 5 \end{bmatrix}$

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Image processing by linear algebra

1. An image is a large matrix of grayscal values, one for each pixel and color
2. When nearby pixels are correlated (not random), the image can be compressed
3. SVD separates any matrix A into rank one pieces (simple pieces) $\mathbf{u}\mathbf{v}^T = (\mathbf{column})(\mathbf{row})$. It is useful to compress image
4. The rows and columns are eigenvectors of $A^T A$ and $A^T A$

Example - Low Rank Images

Consider an image represented by

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

If you send $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ then you need to use 36 numbers (36 pixels - each pixel require 8 bits of information).

Observe that

$$A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Instead of sending all elements in matrix A, we can send a column $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

and a row $\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ which require 12 numbers

$$\textbf{Don't send } A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \qquad \textbf{Send this } A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

With 300 by 300 hundred pixels, 90000 numbers becomes 600

Rank 1 pattern $A = \mathbf{u}\mathbf{v}^T$

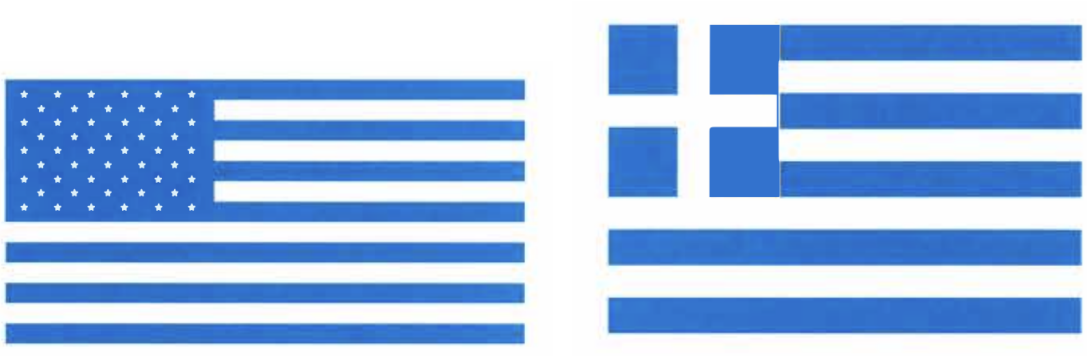
$$\textbf{Don't send } A = \begin{bmatrix} a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \end{bmatrix} \qquad \textbf{Send } A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} a & a & c & c & e & e \end{bmatrix}$$

Rank 2 pattern $A = c_1\mathbf{u}_1\mathbf{v}_1^T + c_2\mathbf{u}_2\mathbf{v}_2^T$ or higher

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ is equal to } A = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

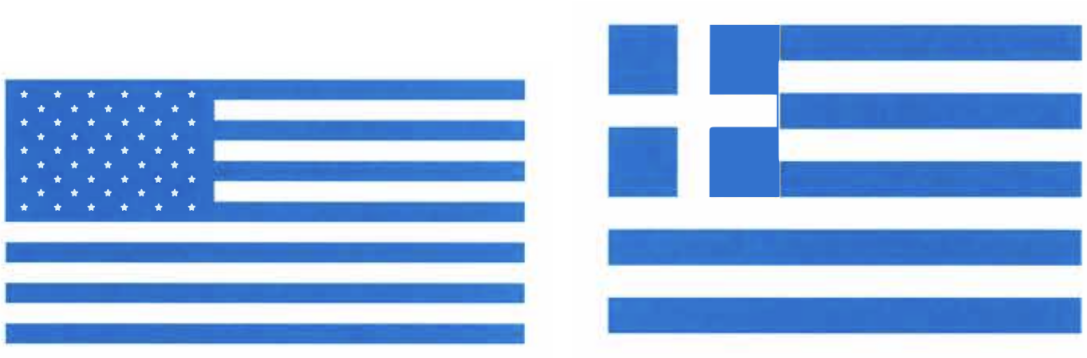
Rank 2 pattern $A = c_1\mathbf{u}_1\mathbf{v}_1^T + c_2\mathbf{u}_2\mathbf{v}_2^T$ or higher

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ is equal to } A = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$



Rank 2 pattern $A = c_1\mathbf{u}_1\mathbf{v}_1^T + c_2\mathbf{u}_2\mathbf{v}_2^T$ or higher

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ is equal to } A = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$



If the rank of A is much higher than 2 (for real images) then A will add up may rank one pieces

$$A = \sum_{i=1}^{n} \sigma_i \mathbf{u}_i \mathbf{v}_i^T \text{ for } n \geq 2$$

We want the small ones such that thay can be discarded with no loss of visual quantity - image compression.

In other words, we wan to decompose an $m \times n$ matrix A into

$$A = U\Sigma V^T$$

where U, V are orthogonal matrices and matrix Σ is diagonal
Starting from $A^T A$ is symmetric, there exists an orthogonal matrix $V = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix}$ such that

$$V(A^T A)V^T = D$$

where D is a diagonal matrix whose diagonal entries $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $A^T A$, that is

$$A^T A\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

Decomposition $A = U\Sigma V^T$ leads to $AV = U\Sigma$ that require $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$.
Hence $AA^T\mathbf{u}_i = \frac{1}{\sigma_i} AA^T A\mathbf{v}_j = \frac{1}{\sigma_i} A\lambda_i \mathbf{v}_i = \frac{\lambda_i}{\sigma_i} A\mathbf{v}_i = \lambda_i \mathbf{u}_i$. That is λ_i is also an eigenvalue of AA^T and $\mathbf{u}_i$ is its corresponding eigenvector.
Remark that $\|A\mathbf{v}_i\|^2 = \lambda_i \|\mathbf{v}_i\|^2 \geq 0 \Rightarrow \lambda_i \geq 0$ and we can verify that $\lambda_i \mathbf{v}_i = A^T A\mathbf{v}_i = \sigma^2 \mathbf{v}_i$. Hence $\sigma_i = \sqrt{\lambda_i}$

Lemma

1. *All eigenvalues of $A^T A$ and AA^T are non-negative*
2. *$A^T A$ and AA^T have the same set of positive eigenvalues $\{\lambda_i\}$*

Definition

The real numbers $\sigma_i = \sqrt{\lambda_i}$ are called **singular values** of the matrix A

Theorem (SVD theorem)

Suppose that A is a matrix of rank r and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ be positive singular values of A then

$$A = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$$

*where $\boldsymbol{u}_i$ and $\boldsymbol{v}_i$ are orthonormal eigenvector corresponding to eigenvalues $\lambda_i = \sigma_i^2$ of $A^T A$ and AA^T, called **left singular vectors** and **right singular vector respectively***

Theorem (SVD theorem (cont))

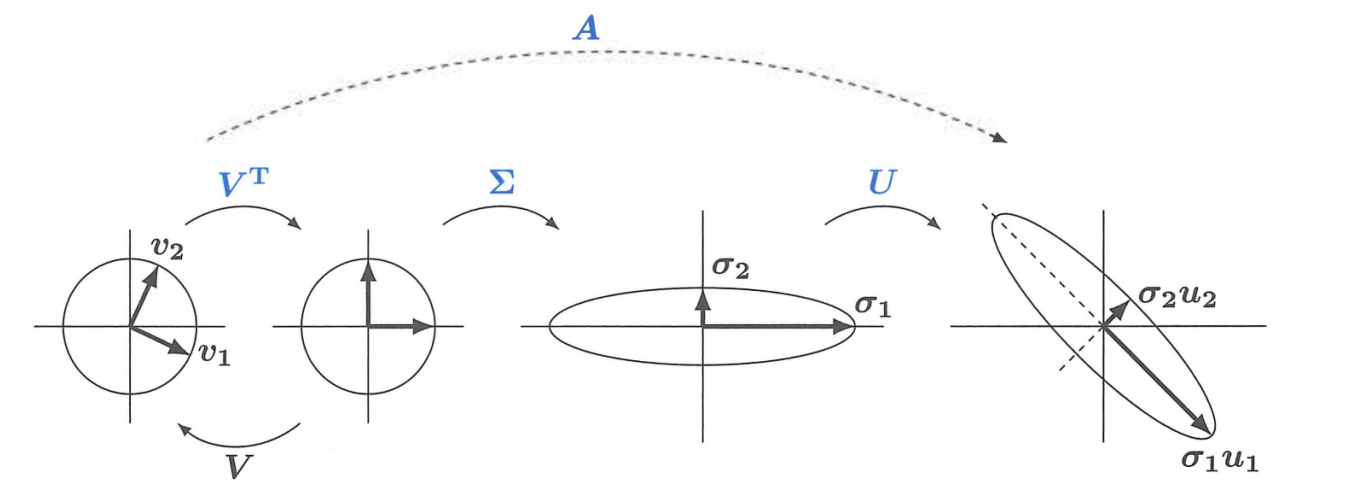
In other words

$$A = U\Sigma V^T$$

where U *and* V *are orthogonal matrix of* A^TA *and* AA^T *and*

$$\Sigma = \begin{bmatrix} diag(\sigma_1, \ldots, \sigma_r) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

in block form, which is called ***singular matrix*** *of* A



Geometric meaning (**rotation**) $\times$ (**stretching**) $\times$ (**rotation**)

SVD algorithm

In order to obtain SVD of A,

1. Compute $A^T A$ and AA^T
2. Find the eigenvalues λ_i of $A^T A$ and then singular values $\sigma_i = \sqrt{\lambda_i}$ of A to create singular matrix Σ
3. Find orthogonal matrix of $A^T A$ and AA^T
4. Decompose $A = U\Sigma V^T$

Example

Find an SVD for $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Solution

First we compute

$$AA^T = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \text{ and } A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Next we find eigenvalues by solving the characteristic polynomial of AA^T

$$\det \begin{bmatrix} 1-x & 1 \\ 1 & 2-x \end{bmatrix} = 0 \leftrightarrow x^2 - 3x + 1 = 0$$

gives eigenvalues $\lambda_1 = \frac{3+\sqrt{5}}{2}$ *and* $\lambda_2 = \frac{30\sqrt{5}}{2}$.
The singular values are $\sigma_1 = \sqrt{\lambda_1} = \frac{\sqrt{5}+1}{2}$, $\sigma_2 = \sqrt{\lambda_2} = \frac{\sqrt{5}-1}{2}$

Solution (cont)

The orthogonal eigenvectors of AA^T and A^TA are

$$\boldsymbol{u}_1 = \frac{1}{\sqrt{1+\sigma_1^2}}\begin{bmatrix}1\\ \sigma_1\end{bmatrix}, \boldsymbol{u}_2 = \frac{1}{\sqrt{1+\sigma_1^2}}\begin{bmatrix}\sigma_1\\ -1\end{bmatrix}, \boldsymbol{v}_1 = \frac{1}{\sqrt{1+\sigma_1^2}}\begin{bmatrix}\sigma_1\\ 1\end{bmatrix}, \boldsymbol{v}_2 = \frac{1}{\sqrt{1+}}$$

Lastly we decompose

$$A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^T + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^T$$

The nearest rank 1 matrix to A will be $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^T$.

Exercise

This flag has rank 2

$$A_{Finland} = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 1 \end{bmatrix}$$

Find a SVD decomposition of $A_{Finland}$ and the nearest rank 1 matrix to A.